

S. No.	Heading	Response
1	Learner Name (For certificate)	ANEAK GOPINATH DONGRE
2	Roll Number	2414104725
3	Name of the Project (Choose from 1–6)	IBM i Incident Insights & Response Assistant
4	Description of the project	IBM i Incident Insights & Response Assistant is an AI-assisted operational analysis application designed to analyze IBM i incident data. The application validates and analyzes incident patterns, generates structured operational findings, and uses Google Gemini AI to provide additional insights. A human review approach is maintained, and Gmail API integration supports sharing AI-generated insights through email.
5	Problem / Opportunity Statement	IBM i operational teams may need to review large volumes of incident information to identify recurring patterns, operational risks, and areas requiring attention. Manual analysis can be time-consuming and may make it difficult to quickly summarize important findings. This project explores how data analysis and AI-assisted insights can support faster operational understanding while maintaining human review and validation.
6	AI Tools / Applications used	Google Gemini AI, Gemini API, Google Cloud Platform, Gmail API, Google OAuth 2.0, Python, Streamlit, Pandas, and Google AI Studio.
7	Potential outcome / benefits from the project	The project can help improve visibility into IBM i incident patterns and reduce the effort required to review operational information. It provides structured, data-driven findings combined with AI-assisted interpretation, helping users identify potential risks and investigation areas. Gmail integration also supports easier communication of generated insights

		while maintaining human review and validation.
8	Access to the project files / data	I can only upload the pdf as a deliverable in assignment.

IBM i Incident Insights & Response Assistant

AI-assisted operational incident analysis, pattern identification, and stakeholder communication



Problem & Opportunity

IBM i operational teams routinely inspect extensive incident logs to isolate critical patterns and operational risks. Manual analysis can be labor-intensive and makes it challenging to rapidly summarize actionable takeaways under tight operational timelines.



Project Objective

Develop an AI-assisted operational analysis application that ingests IBM i incident data, isolates validated operational findings, produces structured Gemini AI-assisted interpretations, and streamlines communication through email distribution.



Key Technologies

A modular cloud-ready architecture built with industry-standard frameworks



Python

Data validation, analytical parsing, and orchestration



Streamlit

Interactive incident dashboard and operator workflow UI



Google Gemini AI

Structured operational insights, hypotheses, and summaries



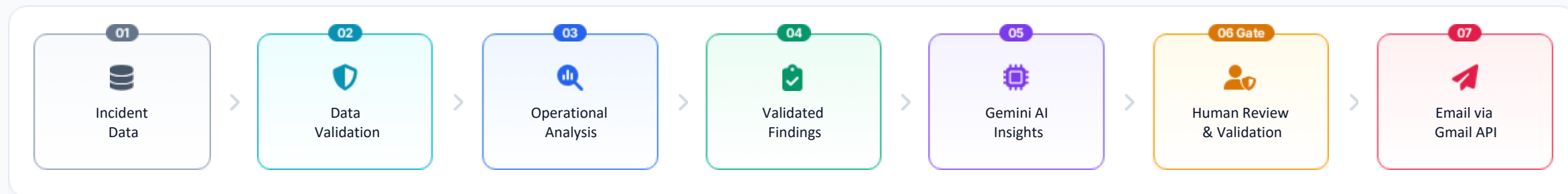
Gmail API & Google OAuth

Secure stakeholder insight dispatch with OAuth 2.0 governance

SYSTEM ARCHITECTURE & GOVERNANCE

Solution Workflow

Deterministic pipeline linked with AI-assisted insights and mandatory human-in-the-loop validation



Architectural Separation of Concerns

The application strictly separates verified, data-driven findings from AI-assisted interpretations. AI-generated risks, hypotheses, and recommendations are advisory and require before operational use **human review and validation**

Core Application Capabilities

- ✓ Incident dashboard & triage
- ✓ Validated operational metrics
- ✓ Gmail-based email sharing
- ✓ Pattern identification
- ✓ AI-generated insights
- ✓ Human sign-off workflow

Key Features & Expected Benefits

Enabling rapid operational visibility while preserving human oversight



Key Application Features

- ✓ Interactive Streamlit UI: Accessible incident analysis dashboard
- ✓ Operational Pattern Analysis: Aggregates repetitive errors and alerts
- ✓ Validated Findings: Deterministic Python/Pandas analysis of operational patterns and metrics
- ✓ Gemini AI Integration: Structured contextual hypotheses and guidance
- ✓ Human Oversight Approach: Mandatory validation before actioning
- ✓ Gmail API Integration: Secure sharing of AI-generated insights through email
- ✓ Secure Credential Isolation: Safe token handling with synthetic data



Expected Operational Benefits

- ✓ Faster Pattern Recognition: Helps reduce the effort required to review incident patterns
- ✓ Improved Risk Visibility: Highlights operational patterns and potential areas requiring investigation
- ✓ Structured AI Analysis: Standardized, repeatable analytical summaries
- ✓ Enhanced Stakeholder Comms: Clear, shareable executive summaries
- ✓ Reusable Prototype: Can be extended with future IBM i and DB2 for i data integration
- ✓ Human Oversight: AI-generated insights require review and validation before operational decisions



Capstone Summary: This project demonstrates how Python, Gemini AI, and API integration can augment operational incident analysis while maintaining strict human oversight of AI-generated recommendations.

IBM i Incident Insights & Response Assistant

Application Screenshots and Demonstration Walkthrough

Project Overview & Architecture

- Domain: IBM i Operational Incident Management
- Analysis Engine: Python and Pandas for deterministic data processing and operational KPI calculation
- AI Engine: Google Gemini for advisory operational insights and risk assessment
- Governance: Human-in-the-loop review model; AI insights are strictly advisory and require operational validation
- Collaboration: Gmail API integration for distributing generated reports

Section 1: Main Incident Dashboard

The Main Incident Dashboard provides an operational overview of the IBM i incident dataset and system performance. It displays high-level KPIs calculated directly from validated incident data:

- Total Incidents: 580
- Critical + High Incidents: 190
- Open Incident Backlog: 59
- Average Resolution Time: 115.8 minutes
- Total Affected Users: 35,132

Deploy

 **IBM i Incident Insight & Response Assistant**

AI-Assisted Operational Incident Analysis

This prototype analyses synthetic IBM i operational incident data to identify patterns, calculate key performance indicators, and generate advisory insights to support human decision-making.

 Responsible AI Notice: AI-generated insights in this application are advisory only. Final operational decisions must be reviewed by an authorised IBM i support professional.

Incident dataset loaded successfully.

Dataset validation completed.

[View Data Validation Results](#)

Incident Overview

A high-level operational snapshot of incident volume, severity, backlog, resolution performance, and business impact, calculated directly from the validated incident dataset.

Total Incidents
580

Critical + High Incidents
190

Open Incident Backlog
59

Avg Resolution Time
115.8 min

Total Affected Users
35,132

KPI Definitions

Incident Pattern Analysis

The charts below highlight incident distribution, system impact, resolution performance, and incident activity trends based on the validated incident dataset.

Visual Pattern Analysis

- Incidents by Severity: Critical (50), High (140), Medium (214), Low (176)
- Incidents by Alert Category: Performance (179), Work Management (109), Storage (100), Spool (93), Network (60), Security (39)
- Incidents by System: IBMI-PROD-01 (289), IBMI-PROD-02 (175), IBMI-DR-01 (72), IBMI-TEST-01 (44)
- Average Resolution Time by Category: Storage (274.7 min), Network (130 min), Performance (105 min), Security (88 min), Work Management (68 min), Spool (18 min)
- Daily Incident Trend: Highlights spikes in incident frequency across June 2026, peaking on June 27 (74 incidents).



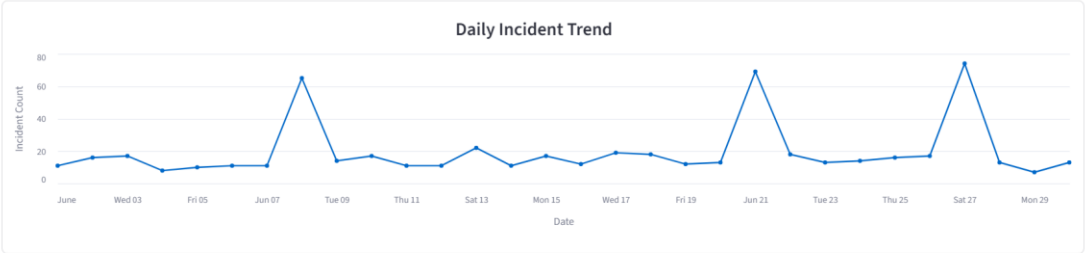


Chart Guide

- Incidents by Severity: Shows the distribution of incidents across severity levels.
- Incidents by Alert Category: Shows which operational alert categories occur most frequently.
- Incidents by System: Shows incident volume across IBM i systems.
- Average Resolution Time: Shows the average time required to resolve incidents by alert category.
- Daily Incident Trend: Shows how incident volume changes over time.

- ## Key Operational Findings (Deterministic Python Logic)
- Alert Pattern: Performance is the most frequent alert category (179 incidents).
 - System Impact: IBMI-PROD-01 recorded the highest incident volume (289 incidents).
 - Resolution Concern: Storage alerts required the longest average resolution time (274.7 minutes).
 - Peak Activity: June 27, 2026 recorded the highest volume with 74 incidents.
 - Critical Incident Risk: 50 critical incidents (8.6% of total volume).
 - Operational Backlog: 59 unresolved open incidents.



Key Operational Findings

The findings below are calculated directly from the validated incident dataset using deterministic Python logic.

Alert Pattern Most Frequent Alert Category: PERFORMANCE Incident Volume: 179 incidents	System Impact Highest Incident Volume: IBMI-PROD-01 Incident Count: 289 incidents
Resolution Concern Longest Average Resolution Time: STORAGE Average Resolution: 274.7 minutes	Peak Activity Peak Incident Day: 27 Jun 2026 Incidents Recorded: 74
Critical Incident Risk Critical Incidents: 50 Percentage of Total: 8.6%	Operational Backlog Current Open Incidents: 59 Metric: Incidents not yet marked as Resolved

Capstone Prototype | Synthetic Data Only | IBM i Incident Insight & Response Assistant

Section 2: Incident Data Explorer

The Incident Data Explorer provides transparent access to the underlying structured incident records supporting the dashboard metrics and AI insights. This ensures the application is anchored in validated operational data rather than ungrounded AI output.

Data Schema & Operational Fields

- Incident ID (e.g., INC0001, INC0002)
- Timestamp and System Identifier (IBMI-PROD-01, IBMI-PROD-02, IBMI-DR-01, IBMI-TEST-01)
- Subsystem (QSYSWRK, QSERVER, QSPL, QINTER, QBATCH)
- Alert Category and IBM i Message ID (e.g., CPI099A, CPI1144, CPI3311, CPD2206)
- Severity, Description, and Resolution Status
- Resolution Duration (minutes), Business Impact, Affected Users, and Assigned Team

Dashboard

Incident Data

AI Insights

Incident Data Explorer

The dataset contains 580 incident records. Select how many records you want to display.

Number of records to display

20

incident_id	timestamp	system_name	subsystem	alert_category	message_id	severity	description	resolution_status	resolution_duration_minutes	business_impact	affected_users	assigned_team	
0	INC0001	2026-06-01 08:14:00	IBMI-PROD-02	QSYSWRK	STORAGE	CPI099A	Critical	System storage limit approaching emergency reserve	Resolved	545	Critical	258	Storage Support
1	INC0002	2026-06-01 08:26:11	IBMI-PROD-01	QSYSWRK	STORAGE	CPI099A	High	System storage limit approaching emergency reserve	Resolved	297	High	118	Storage Support
2	INC0003	2026-06-01 09:37:37	IBMI-DR-01	QSYSWRK	PERFORMANCE	CPI1144	Medium	Machine pool faulting rate critical, adjusting dynamic tuning	Resolved	115	Critical	244	Performance Support
3	INC0004	2026-06-01 11:41:09	IBMI-PROD-02	QSYSWRK	PERFORMANCE	CPI1144	Medium	Machine pool faulting rate critical, adjusting dynamic tuning	Resolved	150	Low	6	Performance Support
4	INC0005	2026-06-01 12:07:59	IBMI-PROD-01	QSPL	SPOOL	CPI3311	Low	Spooled file number limit reached for job log creation	Resolved	13	Low	2	IBM i Operations
5	INC0006	2026-06-01 12:40:24	IBMI-PROD-01	QSERVER	NETWORK	CPI9160	Low	Host server daemon stopped unexpectedly or failed port binding	Resolved	58	Low	2	Network Support
6	INC0007	2026-06-01 14:42:01	IBMI-PROD-01	QSYSWRK	SECURITY	CPD2206	Medium	Unauthorized attempt to modify system values or exit programs	Resolved	130	Medium	45	Security Support
7	INC0008	2026-06-01 14:58:24	IBMI-PROD-01	QINTER	WORK_MGMT	CPF1393	Low	User profile job count limit reached in interactive subsystem	Resolved	55	Low	8	IBM i Operations
8	INC0009	2026-06-01 15:17:36	IBMI-PROD-01	QSYSWRK	PERFORMANCE	CPI1144	Medium	Machine pool faulting rate critical, adjusting dynamic tuning	Resolved	71	Medium	31	Performance Support
9	INC0010	2026-06-01 16:27:51	IBMI-PROD-02	QBATCH	PERFORMANCE	CPI0997	Medium	Subsystem pool paging rate exceeds baseline threshold	Resolved	111	Medium	33	Performance Support
10	INC0011	2026-06-01 16:44:34	IBMI-TEST-01	QSPL	SPOOL	CPI3388	Low	Print device status in intervention required or hung state	Resolved	11	Medium	11	IBM i Operations
11	INC0012	2026-06-02 01:02:41	IBMI-PROD-02	QSERVER	NETWORK	CPD3898	Medium	Enterprise gateway connection timeout during host session negot	Resolved	86	Low	6	Network Support
12	INC0013	2026-06-02 02:15:17	IBMI-TEST-01	QBATCH	WORK_MGMT	CPF1124	Low	Job entered MSGW status awaiting operator response	Resolved	44	Low	3	IBM i Operations

Section 3: AI-Generated Operational Insights

The AI-Generated Operational Insights module allows operators to invoke Google Gemini to evaluate validated findings and generate structured operational analysis.

Advisory Nature & Responsible AI Governance

AI-generated insights are strictly advisory and assist human decision-making. All generated risks, hypotheses, and recommendations require review and validation by an authorized IBM i specialist before operational decisions are taken.

IBM i Incident Insight & Response Assistant

AI-Assisted Operational Incident Analysis

This prototype analyses synthetic IBM i operational incident data to identify patterns, calculate key performance indicators, and generate advisory insights to support human decision-making.

⚠️ Responsible AI Notice: AI-generated insights in this application are advisory only. Final operational decisions must be reviewed by an authorised IBM i support professional.

Incident dataset loaded successfully.

Dataset validation completed.

> 🔍 View Data Validation Results

Dashboard Incident Data **AI Insights**

🔊 AI-Assisted Operational Insights

AI-Generated Insights Notice: Operational metrics and findings are calculated using Python/Pandas from the validated incident dataset. Gemini AI is used to interpret these validated findings and generate hypotheses and recommendations. AI-generated hypotheses require further operational validation and should not be treated as confirmed root causes.

⚡ Generate AI Insights

🗑️ Clear Insights



The user initiates the AI analysis by selecting the Generate AI Insights option.

⚡ Generate AI Insights

🗑️ Clear Insights

🔄 Gemini is analyzing the operational findings...

Core Output Modules

- **Executive Summary:** Synthesizes key operational patterns across system drivers, alert categories, resolution bottlenecks, and backlog metrics.
- **Most Important Operational Risk:** Details critical risks, such as prolonged storage incident resolution (averaging 274.7 minutes) on primary production partition IBMI-PROD-01.
- **Possible Reasons Requiring Investigation:** Hypothesizes potential causes requiring manual validation, such as manual ASP disk cleanup, batch workload contention, and peak day scheduled runs.
- **Recommended Priority Actions:**
 - **Immediate Actions:** Triage the 59-incident backlog; establish active storage alerting thresholds; inspect June 27 historical job logs.
 - **Preventive Actions:** Analyze Collection Services data for recurring bottlenecks; optimize storage runbooks; evaluate batch and backup scheduling windows.

⚡ Generate AI Insights

🗑️ Clear Insights

AI operational insights generated successfully.

📊 AI Operational Analysis

🔊 AI-Generated Operational Insights

The following analysis is generated from validated operational findings. AI-generated risks, hypotheses, and recommendations require operational review and validation.

1. Executive Summary

Based on the validated operational findings, the following key patterns and operational concerns have been identified:

- **Primary System Driver:** `IBMI-PROD-01` is the main contributor to operational volume, accounting for 289 incidents.
- **Dominant Alert Category:** `PERFORMANCE` is the most frequent alert category, with 179 incidents recorded.
- **Resolution Bottleneck:** `STORAGE` alerts present the longest average resolution time, taking an average of 274.7 minutes to resolve.
- **Peak Activity:** A significant spike occurred on 27 Jun 2026, with 74 incidents recorded on that day.
- **Severity and Backlog:** There are 50 validated critical incidents (representing 8.6% of the total). The operations team is currently carrying an active backlog of 59 open incidents.

Further investigation is required to determine the underlying relationships between these findings, as the dataset does not provide explicit root causes.

2. Most Important Operational Risk

Observation

- STORAGE alerts take an average of 274.7 minutes to resolve.
- PERFORMANCE is the most frequent alert category with 179 incidents.
- The production partition IBM1-PROD-01 has the highest incident volume (289 incidents).

Risk

If storage-related alerts remain unresolved for over 4.5 hours on average, there is a risk of critical system degradation or potential application disruption on the production system (IBM1-PROD-01) if storage utilization approaches capacity limits. Additionally, the high frequency of performance alerts on the primary production partition risks impacting business operations and SLA compliance.

3. Possible Reasons Requiring Investigation

The following areas are hypotheses based on the observed patterns and require validation with IBM i operational data:

- Storage Resolution Procedures:** The high average resolution time for storage alerts (274.7 minutes) may indicate that disk space analysis, ASP (Auxiliary Storage Pool) cleanup, or temporary storage reclamation procedures are manual, complex, or lack automated alerting thresholds. This area could be investigated to determine why these alerts require extended resolution times.
- Production Workload and Performance:** The concentration of performance alerts (179) and general incident volume on IBM1-PROD-01 (289) may indicate workload constraints, resource contention, or batch processing issues. This workload distribution could be reviewed using system performance metrics.
- Peak Day Activity (27 Jun 2026):** The spike of 74 incidents on June 27, 2026, may correspond to scheduled processing, a month-end/quarter-end batch run, or a specific system maintenance window. This peak requires validation against scheduled job logs and change management records for that date.

4. Recommended Priority Actions

These recommendations are based on the observed patterns and are not confirmed root-cause solutions.

Immediate Actions

- Triage Current Backlog:** It is recommended to review and prioritize the 59 open incidents in the backlog, with immediate focus on any open critical, performance, or storage incidents associated with IBM1-PROD-01.
- Establish Storage Alert Protocols:** Because storage alerts take the longest to resolve, it may be worth reviewing the active storage allocation and temporary storage usage on IBM1-PROD-01 to ensure no critical thresholds are currently breached.
- Investigate Peak Day Event:** It is recommended to inspect historical system logs, job queues, and scheduled tasks for June 27, 2026, to identify what may have triggered the 74 incidents.

Preventive Actions

- Analyze Performance Trends:** To address the high volume of performance incidents, it is recommended to analyze Collection Services data on IBM1-PROD-01 to identify recurring resource bottlenecks, heavy batch jobs, or subsystem constraints.
- Optimize Storage Incident Runbooks:** To reduce the 274.7-minute average resolution time, it is recommended to review and update operational runbooks for storage-related alerts, potentially exploring automated disk cleanup scripts or earlier warning thresholds.
- Review Backup and Maintenance Windows:** It may be beneficial to evaluate whether scheduled maintenance, backup operations, or batch processing schedules can be adjusted to minimize performance impacts on the production environment.

Download AI Insights Report

Section 4: Download and Email Insights

Once operational insights are generated, the application provides built-in sharing and dissemination options:

- Report Export:** Users can download the complete AI Insights Report locally for documentation and archival.
- Gmail Integration:** Securely email the structured findings and recommendations directly to operations team members or management stakeholders.

Send AI Insights via Email

Send the generated AI operational insights to an operations team member or stakeholder.

Recipient Email Address

name@example.com

Send AI Insights